Incompressible fluids I
Darcy Flow and Raviart–Thomas mixed FEM

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REFERENCES

Larson–Bengzon, *The Finite Element Method*, Springer, 2013.

Quarteroni, *Numerical Models for Differential Problems*, Springer, 2017.

Boffi–Brezzi–Fortin, *Mixed Finite Element Methods and Applications*, Springer, 2013.

Girault–Raviart, *Finite Element Methods for Navier-Stokes Equations*, Springer, 1986.

Conservation of mass

A fluid occupying the volume Ω has density ρ and velocity u.

Over a control volume $\omega \subset \Omega$,

$$\frac{d}{dt}\int_{\omega} \rho \, dx + \int_{\partial\omega} \rho u \cdot n \, ds = 0.$$

Using the divergence theorem,

$$\int_{\omega} \big(\partial_t \rho + \nabla \cdot (\rho u)\big) \, dx = 0.$$

Since ω is arbitrary,

$$\partial_t \rho + \nabla \cdot (\rho u) = 0.$$

In particular, if $\rho \equiv \text{const}$, then

$$\nabla \cdot u = 0,$$

which is the incompressibility constraint.

Darcy's law (1856)

Darcy's law is a phenomenological constitutive equation for flow through porous media. At constant elevation, the discharge rate satisfies

$$Q = -\frac{kA}{\mu L}(p_b - p_a),$$

where

- Q is the discharge,
- μ the viscosity,
- k the intrinsic permeability,
- A the cross-sectional area,
- L the flow length.

Pointwise, for the *homogenised* Darcy velocity u, this becomes

$$u = -K\nabla p.$$

with $K := k/\mu$. Equivalently,

$$K^{-1}u + \nabla p = 0.$$

Darcy system and its connction with Poisson

Combine Darcy's law with conservation of mass:

$$-\nabla \cdot u = f \qquad \text{in } \Omega.$$

With a pressure boundary condition,

$$p = g \qquad \text{on } \partial\Omega,$$

we obtain the mixed Darcy system

$$\begin{cases} K^{-1}u + \nabla p = 0 & \text{in } \Omega, \\ -\nabla \cdot u = f & \text{in } \Omega, \\ p = g & \text{on } \partial\Omega. \end{cases}$$

Eliminating u gives

$$-\nabla \cdot (K \nabla p) = f,$$

so Darcy is a mixed formulation of Poisson.

Key point. Mixed formulation $\Rightarrow$ direct access to both pressure and flux.

Weak mixed formulation

The natural flux space is

$$H(\mathrm{div};\Omega) := \{v \in [L^2(\Omega)]^d : \nabla \cdot v \in L^2(\Omega)\}.$$

Testing the first equation with $v \in H(\mathrm{div};\Omega)$ and integrating by parts,

$$\int_\Omega K^{-1} u \cdot v \, dx - \int_\Omega p \, \nabla \cdot v \, dx = -\int_{\partial\Omega} g \, v \cdot n \, ds.$$

Testing the second equation with $q \in L^2(\Omega)$,

$$-\int_\Omega q \, \nabla \cdot u \, dx = \int_\Omega fq \, dx.$$

Hence: find $(u, p) \in H(\mathrm{div};\Omega) \times L^2(\Omega)$ such that

$$\begin{aligned} a(u, v) - b(v, p) &= -(g, v \cdot n)_{\partial\Omega} \quad && \forall v, \\ -b(u, q) &= (f, q)_\Omega && \forall q, \end{aligned}$$

with

$$a(u, v) := (K^{-1} u, v)_\Omega, \qquad b(v, q) := (\nabla \cdot v, q)_\Omega.$$

Saddle-point structure

Weak mixed formulation: find $(u, p) \in H(\mathrm{div}; \Omega) \times L^2(\Omega)$ such that

$$\begin{aligned} a(u, v) - b(v, p) &= -(g, v \cdot n)_{\partial\Omega} \quad && \forall v, \\ -b(u, q) &= (f, q)_\Omega && \forall q, \end{aligned}$$

This is a saddle-point problem for the complete form

$$\mathcal{A}((u, p), (v, q)) = a(u, v) - b(v, p) - b(u, q).$$

The correct abstract tool is the Banach–Nečas–Babuška theorem (key stability mechanism is the Babuška–Brezzi inf–sup condition, instead of the simpler coercivity). **More difficult!**

Equivalence with constraint minimisation

Recall that the primal problem is equivalent (if $g = 0$) to a minimisation problem in $H^1_0(\Omega)$.

Instead, the mixed formulation is equivalent to the constraint minimisation problem:

$$\text{Find } u \in H(\mathrm{div};\Omega) \, : \, u = \operatorname*{argmin}_{v \in H(\mathrm{div};\Omega)} \int_\Omega K^{-1} u \cdot v + \int_{\partial\Omega} g v \cdot n,$$

subject to the linear constraint

$$b(v,q) = -(f,q) \qquad \forall q \in L^2(\Omega).$$

Why Raviart–Thomas spaces?

We want to discretize Darcy conformingly in $H(\mathrm{div};\Omega)$...

Lemma (Characterization of $H(\mathrm{div};\Omega)$)

A function $v \in H(\mathrm{div};\mathcal{T}_h) \cap W^{1,1}(\mathcal{T}_h)$ belongs to $H(\mathrm{div};\Omega)$ iif

$$[\![v]\!] \cdot n_F = 0 \qquad \forall F \in \mathcal{F}_h^i.$$

So that the normal component needs to be continuous across element interfaces!

This is exactly what Raviart–Thomas spaces provide for $\mathcal{T}_h$.

For a simplex K and degree $k \geq 0$,

$$RT_k(K) := [\mathbb{P}_k(K)]^d + x\,\mathbb{P}_k(K).$$

Thus every $v \in RT_k(K)$ has the form

$$v = w + xp, \qquad w \in [\mathbb{P}_k(K)]^d, \quad p \in \mathbb{P}_k(K).$$

Main properties of $RT_k(K)$

Lemma (Properties of RT)

- *for every face $F \subset \partial K$, $v \cdot n_F \in \mathbb{P}_k(F)$,*
- $\nabla \cdot RT_k(K) = \mathbb{P}_k(K)$.

Proof. If $v = w + xp$, then on a face F,

$$v \cdot n_F = w \cdot n_F + (x \cdot n_F)p.$$

Since $x \cdot n_F$ is constant on F, one gets $v \cdot n_F \in \mathbb{P}_k(F)$.

That $\nabla \cdot v \in \mathbb{P}_k(K)$ is obvious. Surjectivity follows from the fact that $\nabla \cdot (xp) = 0$, $p \in \mathbb{P}_k(K)$, implies $p = 0$.

The local Raviart–Thomas interpolant and DoF

There exists a unique local interpolant

$$I_K^{RT} : [H^1(K)]^d \to RT_k(K)$$

characterized by the moments of the normal trace on faces (DoF):

$$\int_F (I_K^{RT} v \cdot n_F) p_k \, ds = \int_F (v \cdot n_F) p_k \, ds \qquad \forall p_k \in \mathbb{P}_k(F), \ \forall F \subset \partial K,$$

and, for $k \geq 1$, by the internal moments

$$\int_K I_K^{RT} v \, p_{k-1} \, dx = \int_K v \, p_{k-1} \, dx \qquad \forall p_{k-1} \in [P_{k-1}(K)]^d.$$

Lemma (Interpolation error estimate)

If $v \in [H^m(K)]^d$, for some $1 \leq m \leq k+1$ then

$$\|v - I_K^{RT} v\|_{L^2(K)} \leq C h_K^m |v|_{H^m(K)}.$$

Note: the interpolant is defined only requiring $v \in [H^1(K)]^d$.

Global $H(\mathrm{div})$-conforming RT space

Define the global space

$$RT_k := \{v \in H(\mathrm{div};\Omega) : v|_K \in RT_k(K)\ \forall K \in \mathcal{T}_h\}.$$

The global interpolation operator

$$I_h^{RT} : H(\mathrm{div};\Omega) \cap \prod_{K\in\mathcal{T}_h} [H^1(K)]^d \to RT_k,$$

is defined elementwise:

$$I_h^{RT} v|_K := I_K^{RT} v.$$

Because the degrees of freedom are (also) based on the normal trace, the resulting interpolant has continuous normal component across faces, hence indeed

$$I_h^{RT} v \in RT_k.$$

Commuting property

A key result is that

$$\int_\Omega \nabla \cdot (v - I_h^{RT} v)\, q_h \, dx = 0 \qquad \forall q_h \in Q_h,$$

where Q_h is the dG space of degree k, hence

$$Q_h := \{q_h \in L^2(\Omega) : q_h|_K \in \mathbb{P}_k(K)\ \forall K \in \mathcal{T}_h\}.$$

Moreover,

$$\nabla \cdot RT_k = Q_h.$$

Thus we have the commuting property (for Π_h the L^2- projection.)

$$\nabla \cdot I_h^{RT} v = \Pi_h(\nabla \cdot v)$$

This is the key structural fact behind the stability and error analysis of the RT method[1].

[1] RT interpolation preserves the divergence of the vector field in the discrete pressure space (rather than the filed itself)— exactly what mixed Darcy needs.

The commuting property of the RT interpolant

Interpretation: Interpolation and divergence commuting diagram:

$$
\begin{array}{ccc}
H(\mathrm{div};\Omega) & \xrightarrow{I_h^{RT}} & RT_k \\
\nabla\cdot \big\downarrow & & \big\downarrow \nabla\cdot \\
L^2(\Omega) & \xrightarrow[\Pi_h]{} & Q_h
\end{array}
$$

So the RT interpolant preserves the divergence *in the discrete pressure space*.

Consequence:

$$b(v - I_h^{RT} v, q_h) = 0 \qquad \forall q_h \in Q_h,$$

which is exactly the Fortin-type property behind the discrete inf–sup condition (see later).

Raviart–Thomas mixed FEM

Choose

$$U_h := RT_k, \qquad Q_h := \{q_h \in L^2(\Omega) : q_h|_K \in \mathbb{P}_k(K)\}.$$

The discrete mixed problem is: find $(u_h, p_h) \in U_h \times Q_h$ such that

$$\begin{aligned} a(u_h, v_h) - b(v_h, p_h) &= -(g, v_h \cdot n)_{\partial\Omega} && \forall v_h \in U_h, \\ -b(u_h, q_h) &= (f, q_h)_\Omega && \forall q_h \in Q_h. \end{aligned}$$

This is the Raviart–Thomas mixed finite element method for Darcy / mixed Poisson.

Velocity error estimate

Let (u, p) solve the mixed problem and let (u_h, p_h) be the RT solution. Using

$$\|u - u_h\|_{L^2(\Omega)} \le \|u - I_h^{RT} u\|_{L^2(\Omega)} + \|I_h^{RT} u - u_h\|_{L^2(\Omega)},$$

and the commuting property,

$$\|u - u_h\|_{L^2(\Omega)} \le C\Big(1 + \|K\|_{L^\infty}\|K^{-1}\|_{L^\infty}\Big)\|u - I_h^{RT} u\|_{L^2(\Omega)}.$$

Theorem (Velocity error bound)

If $u \in [H^m(\Omega)]^d$ with $1 \le m \le k+1$,

$$\|u - u_h\|_{L^2(\Omega)} \le Ch^m |u|_{H^m(\Omega)}.$$

Crucial point: the velocity error decouples from the pressure error!!

Pressure estimate

For the pressure it holds

$$\|p - p_h\|_{L^2(\Omega)} \leq C\Big(\|p - \Pi_h p\|_{L^2(\Omega)} + \|u - I_h^{RT} u\|_{L^2(\Omega)}\Big),$$

Hence, under suitable regularity assumptions,

$$\|p - p_h\|_{L^2(\Omega)} \leq Ch^m\big(|u|_{H^m(\Omega)} + |p|_{H^m(\Omega)}\big).$$

(On convex domains one can gain one extra order for p by duality.)